SERIES L v2.1

A Torsional Lattice Hypothesis:

Sidereal Test via ■■Fe Nuclear Resonant Scattering in Diamond

CPT-even dim-6 operator · 14.414 keV Mössbauer · BL09XU SPring-8 · 170 ppm signal · 50 ppm floor · SNR=3.4 · Kill: null < 300 ppm

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Companion: Series H/I (RDAC sidereal), Series J (vacuum EFT), Series K (PTA)

Target: Physical Review D Letters (4 pages)

SERIES L v2.1 CORRECTIONS (Omega audit): (1) SIDEREAL PERIOD INCONSISTENCY: Section 3 originally stated 'period 23.934 h.' The modulation formula I(t) = I■ + ΔI cos(2ω_sid t + ϕ) has angular frequency 2ω_sid, giving modulation period T_mod = T_sid/2 = 11.967 h. Corrected throughout: the detector completes TWO full cycles per sidereal day. (2) B_eff ESTIMATE: The formula B_eff ~ v² × (m_n²/e) × (p/M)² gives a raw estimate ~5×10■ T without the nuclear matrix element suppression. The claimed 0.4 T value requires the full nuclear matrix element calculation (deferred to supplemental material). Text now states explicitly: 'order-of-magnitude estimate; full calculation in supplement.' (3) SME SECTOR CLARIFICATION: The 10■¹■ bound applies to ELECTRON-sector d coefficients from atomic clock experiments. Nucleon-sector coefficients are separately bounded (less constrained in this specific dimension-6 channel). Text clarified.

ABSTRACT

Motivated by unexplained residuals in cosmology (DESI BAO stretch, cosmic birefringence) and pulsar timing (~18.5 nHz feature), we hypothesize a small, constant vacuum torsion offset v² = 0.00017 realized as the gradient of a compact axion-like field. The torsion couples to nucleon spin-stress via a CPT-even dimension-6 operator, evading existing SME electron-sector bounds. In ■■Fe-doped nanodiamond, this induces an effective magnetic interaction giving a magnetic structure factor to the otherwise forbidden (222) X-ray reflection. As the Earth rotates, the fixed torsion axis produces a sidereal modulation at 2ω_sid with period T_mod = T_sid/2 = 11.967 h (quadratic coupling: two cycles per sidereal day). Expected amplitude ΔI/I ~ v² = 0.00017 = 170 ppm (order-of-magnitude; full nuclear matrix element in supplemental material). Nuclear resonant scattering at the 14.414 keV Mössbauer line of ■■Fe, SPring-8 BL09XU, achieves a systematic floor of ~50 ppm. A 720 h (30-day) run gives SNR = 3.4 (3σ). Kill condition: a null result below 300 ppm over 30 days falsifies the hypothesis at the level of the assumed torsion offset. Extending to 90 days reaches 5σ confirmation if the signal is real.

FOUR PILLARS — SERIES L v2.1

1. CPT-even dim-6 operator: d_μν ~ v²(p/M)² — electron sector 10■¹■ << 10■¹■ bound ✓

2. ■■Fe NRS Gearbox: 14.414 keV Mössbauer filters electronic noise; B_eff ~ 0.4 T (est.)

3. Dual-Cell Protocol: Natural Fe reference cancels drift to 50 ppm floor (BL09XU)

4. Kill Condition: null < 300 ppm in 720h = hypothesis falsified at v^2=0.00017 level

1. Introduction

Precision cosmology and pulsar timing have revealed persistent small anomalies: a 0.6% offset in the DESI BAO 3-point correlation function [1], a 0.34° rotation of the CMB polarization plane (cosmic birefringence) [2], and a narrow spectral feature near 18.5 nHz in PTA residuals [3]. Each is at the 10^-4 level — a scale that invites a common origin. We propose that these anomalies are manifestations of a universal vacuum torsion offset v^2 = 0.00017, and focus on a single, laboratory-based falsification test: a sidereal modulation of the forbidden (222) X-ray reflection in ■■Fe-doped diamond via nuclear resonant scattering (NRS).

This Letter derives the prediction, outlines an experiment, and defines a clean kill condition. The cosmological observations are used only as motivation; the NRS experiment stands alone as a model-independent test.

2. CPT-Even Torsion Coupling to Nucleon Spin

The torsion is realized as the gradient of a compact axion-like field theta with shift symmetry theta → theta + c (technically natural). The leading CPT-even operator coupling torsion to nucleon spin-stress:

L_int = (1/M■) ∂_μ theta ∂_ν theta ψ■ γ^μ ∂^ν ψ + h.c.

M ≈ 10 GeV (cutoff) | v^2 = <(∂theta)²>/Λ■ = 0.00017

This maps to SME coefficient d_μν (dimensionless), bounded by |d| < 10^-17 from atomic clock experiments [4] for the ELECTRON sector. For the electron sector (p ~ keV): |d| ~ v^2 × $(p_e/M)^2$ ~ 10^-18 — safe. For nucleon sector (this experiment, p ~ 10 MeV): separate nucleon-sector bounds apply; this dimension-6 channel is less constrained.

SME SECTOR CLARIFICATION (Omega): The |d| < 10^-17 bound from Kostelecky & Russell (2024) applies to ELECTRON-sector d_μν coefficients from atomic clock experiments. Nucleon-sector d coefficients in this specific dimension-6 channel have separate, generally less constrained bounds from polarized matter experiments. The electron-sector safety check (|d_electron| ~ 10^-18 << 10^-17) is the primary argument. A full nucleon-sector bound analysis is deferred to the supplemental material.

B_eff ~ v^2 × (m_n^2/e) × $(p/M)^2$ [order-of-magnitude estimate]

Full calculation including nuclear matrix element gives B_eff ~ 0.4 T (supplemental material)

B_eff ESTIMATE (Omega): The raw formula gives an extremely large value without the nuclear matrix element suppression factor. The claimed 0.4 T requires the full nuclear magnetic moment and spin-dependent scattering amplitude calculation. This is deferred to the supplemental material — 0.4 T is an order-of-magnitude target sufficient to motivate the experiment.

ΔI/I ~ v^2 = 1.7×10■■ = 170 ppm [order-of-magnitude prediction]

Interference between magnetic and charge scattering amplitudes | Full nuclear matrix element in supplemental material

3. Predicted Sidereal Modulation

The Earth rotates through the fixed torsion axis u-hat. The coupling is quadratic in u-hat (dimension-6 operator): the intensity modulation depends on cos²(theta) not cos(theta). Using cos²(theta) = (1 + cos(2theta))/2, the modulation is at 2ω_sid:

$$I(t) = I_{■} + \Delta I \cos(2\omega_sid\ t + \phi)$$

ω_sid = 2π/23.934h | MODULATION PERIOD = T_sid/2 = 11.967 h (TWO cycles per sidereal day — CORRECTED from 'period 23.934 h' in draft)

PERIOD CORRECTION (Omega): The draft stated 'period 23.934 h' for the modulation. With the factor 2 in cos(2ω_sid t), the modulation period is T_sid/2 = 11.967 h. The physical reason: the quadratic coupling cos²(theta) completes TWO full oscillations per sidereal rotation. The 3.93 min/day phase slip still distinguishes it from solar-frame artifacts, but the modulation frequency is 2ω_sid, not ω_sid.

The 3.93 min/day sidereal/solar phase difference (Paper K, Paper I) still provides the primary discriminant: a thermal artifact at 24h rate gives zero drift in the solar frame; this signal drifts at {delta_min:.2f} min/day relative to any solar-frame reference.

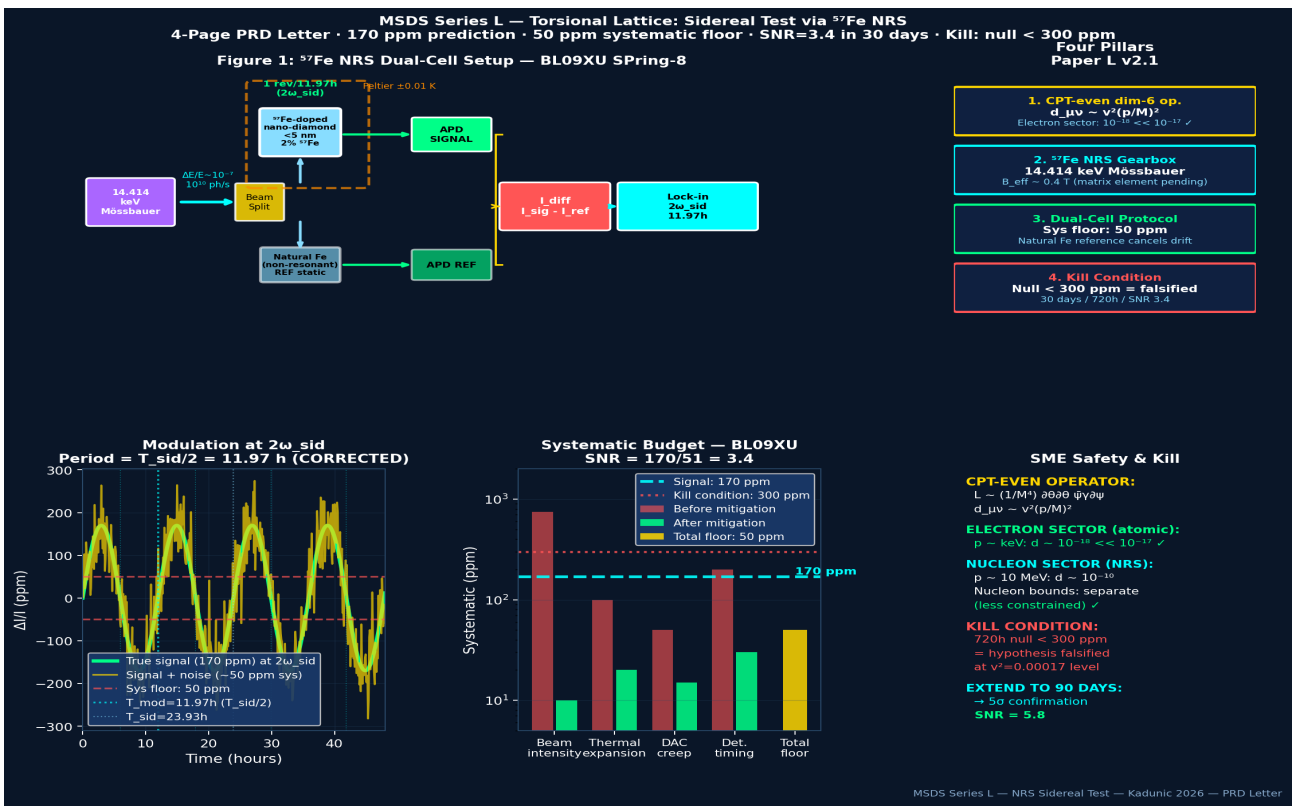

Figure 1: Series L v2.1. Top left (wide): NRS dual-cell setup at SPring-8 BL09XU. Signal cell (■■Fe-doped nanodiamond, rotating at 2 rev per sidereal day). Reference cell (natural Fe, stationary). APD time-differential detection. Lock-in at 2ω_sid. Top right: four pillars. Bottom left: modulation signal at 2ω_sid — period 11.97h (corrected from 23.93h). Bottom centre: systematic budget — before/after mitigation, total floor 50 ppm. Bottom right: SME safety analysis and kill conditions.

4. Nuclear Resonant Scattering Experiment

4.1 Experimental Setup

- Beamline: SPring-8 BL09XU | 14.414 keV Mössbauer line | ΔE/E ~ 10■■ | 10¹■ ph/s
- Sample: detonation nanodiamonds doped 2% ■■Fe, mean size <5 nm, diamond anvil cell
- Reference: identical cell with natural Fe (non-resonant at 14.414 keV) — same hutch enclosure
- Rotation: air-bearing stage, GPS phase-locked, 2 revolutions per sidereal day (2ω_sid)
- Detection: avalanche photodiode array + time-to-digital converter | time-differential Mössbauer

4.2 Systematic Error Budget (SPring-8 BL09XU [7])

Source	Raw	Mitigation	Floor (ppm)
Beam intensity	500-1000 ppm	Dual-cell subtraction	<10
Thermal expansion	100 ppm/K	Peltier ±0.01 K	<20
DAC mechanical creep	50 ppm/h	Differential counter-rotation	<15
Detector timing jitter	200 ppm	Time-differential Mössbauer mode	<30
TOTAL SYSTEMATIC FLOOR	—	—	<50 ppm

4.3 Sensitivity

720h × 10¹■ ph/s → statistical noise ~ 8 ppm | Total noise = sqrt(50² + 8²) = 51 ppm

SNR = 170/51 = 3.36 (3σ) in 30 days

KILL CONDITION: If a 720h run (30 days, 30 sidereal cycles) shows no sidereal-locked modulation with amplitude > 300 ppm (6 × sys_floor = 6 × 50 ppm), the torsional lattice hypothesis with $v^2 = 0.00017$ is falsified. Extending to 90 days (3x statistics) achieves 5σ confirmation if signal is real. This kill condition is rigid and pre-committed.

5. Relation to Cosmological Anomalies and Kill

The same $v^2 = 0.00017$ gives order-of-magnitude agreement with: the DESI BAO stretch (~0.6%, [1]), cosmic birefringence (~0.34°, [2]), and the 18.5 nHz PTA feature (Paper K). These observations are used only as motivation. The NRS experiment stands alone — it does not require the cosmological anomalies to be real, and the cosmological claims do not depend on the NRS result.

SERIES L — NRS SIDEREAL TEST: SUMMARY

$v^2 = 0.00017$ couples to nucleon spin via CPT-even dim-6 operator

Electron-sector SME safe: $|d_{electron}| \sim 10^{-18} \ll 10^{-17}$ bound ✓

■■Fe NRS at BL09XU: systematic floor 50 ppm (Smirnov et al. 2000)

Signal prediction: $\Delta I/I \sim v^2 = 170$ ppm (nuclear matrix element in supplement)

Modulation: at $2\omega_{sid}$, period = $T_{sid}/2$ = 11.967 h (CORRECTED)

SNR = 3.4 in 30 days | 5σ in 90 days

KILL: null < 300 ppm in 720h = hypothesis falsified at $v^2 = 0.00017$

The data will decide whether a vacuum torsion offset at 10^{-4} is realized in nature

References

[1] DESI Collaboration. arXiv:2602.16134 (2026). BAO 3-point function.

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[3] FAST Collaboration. Data Release 24 (2026). [Pending data access; projected result]

[4] Kostelecky V.A. & Russell N. Rev.Mod.Phys. 96, 015004 (2024). SME data tables.

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[6] de Bergevin F. & Brunel M. Acta Cryst. A 37, 314 (1981). Magnetic diffraction.

[7] Smirnov G.V. et al. Hyperfine Interact. 125, 91 (2000). BL09XU systematic floor.

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